

Motivation Letter

Programme: National PhD in Artificial Intelligence — Security & Cybersecurity area (Sapienza Rome + CNR consortium) **Applicant:** Nauval Zulfikar

Dear Members of the Selection Committee,

I am applying to the Security & Cybersecurity area of the National PhD in Artificial Intelligence because it is the place where the two halves of my work finally meet: building transformer-based models, and being accountable for their behaviour once they are deployed.

My interest in deployed-model security did not begin in a research lab — it began at a licensed bank. As Lead Data Scientist at Bank Muamalat Indonesia, I was responsible for keeping NLP and risk-scoring models trustworthy after they went live: running regular validation cycles, monitoring model staleness, and producing documentation that had to satisfy regulatory and audit frameworks. That experience changed how I think about AI. A model that passes its test set can still fail quietly in production, and in a regulated, adversarial environment that quiet failure is the real risk. I learned to treat the deployed lifetime of a model — not its training run — as the place where security and trust are actually decided.

I then deepened the technical side. My MSc at Aston University (Business Analytics, First Class) centred on a dissertation in which I fine-tuned the DeBERTa-v3 transformer end-to-end on multi-tier supplier review data and evaluated its emergent biases against literature baselines. More recently I built an end-to-end retrieval-augmented generation system around gpt-4o-mini with a fully traceable, rule-based fallback — designed deliberately to be reproducible and ablatable. Working hands-on with fine-tuned transformers and LLM agents made the attack surface vivid to me: poisoned fine-tuning data, hidden backdoors that survive ordinary evaluation, adversarial inputs, and agents that act through tools. These are precisely the threats this area studies.

My proposed project joins these threads: detecting poisoning and backdoors in fine-tuned transformers using only the model and its inference behaviour, and turning that detection into an auditable, reproducible control for deployed and agentic LLMs. It is, in effect, my bank-era governance instinct rebuilt on a rigorous security foundation.

I am drawn to this programme specifically because it is a national effort that takes the *cybersecurity* of AI seriously — with cyber slots backed by the national cyber agency and the combined research strength of Sapienza and the CNR consortium. That is the right environment to do work that is both technically sound and operationally real. My MSc was taught and assessed in English, and I hold IELTS 7.0, so I can contribute fully from day one.

Thank you for considering my application. I would be honoured to pursue this research within your area.

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